

Computer Science & IT

Discrete Mathematics



Mathematical Logic

Lecture No. 04



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Recap of Previous Lecture



✓ Topic

Practice questions

✓ Topic

Proof by contradiction

✓ Topic

Conditional proof

Topics to be Covered



- ✓ Topic Predicate logic
- ✓ Topic Quantifiers
- ✓ Topic Scope of quantifiers





Predicate Logic

(First order predicate logic)



Topic : Predicate

It is part of a sentence or clause stating something about the subject.

eg:

Ram is a politician

Subject

Predicate

← Proposition
it may be true
or
it may be false

Let, 'p' denotes "is a politician"

$\underbrace{P(\text{Ram})}_{\text{atomic}} : \text{Ram is a politician}$

predicate 'p'
is applied over
the subject Ram

P : is a politician
 S : is a sportsman

$P(x)$: x is a politician
 $S(y)$: y is a sportsman

x & y may be any person
from the
Set of people under consideration

it is called domain
(or)
Universe of discourse

$P(x) \rightarrow S(x)$: if $P(x)$ then $S(x)$

: if x is a politician, then x is a sportsman

it may be true
or it may be false

① if ' x ' is not a politician, then implication will return true, does not matter whether x is a sportsman or not.

② If x is a politician, then x must be sportsman as well for implication to return true.

③ If x is a politician but not a sportsman, then implication will return false

P : is a politician
 S : is a sportsman

① $P(x) \rightarrow S(y)$: if x is a politician, then y is a sportsman

② $P(x) \vee S(y)$: x is a politician or y is a sportsman

③ $P(x) \wedge S(x)$: x is politician cum sportsman

④ $P(x) \leftrightarrow S(y)$: x is a politician if and only if
 y is a sportsman



Topic : Predicate with multiple subjects

Let 'F' denotes "is a friend of"

then ' $\sim F$ ' : "is not a friend of"

$F(x, y)$: x is a friend of y

$F(\text{Shivani}, \text{Shweta})$: Shivani is a friend of Shweta

↑
'F' will either return True
or it will return False

Let G : is greater than

$G(a, b)$: a is greater than b
 $\{ \text{may be true or may be false} \}$

eg. $G(2, 3)$: is false

$G(5, 5)$: is false

$G(5, 3)$: True

eg. $F(x, y, t)$: person 'x' can fool person 'y' at time 't'

$F(\text{Ram}, \text{Shyam}, 10:00 \text{ AM})$



Topic : Quantifiers



- In predicate logic, predicates are used alongside quantifiers to express the extent to which a predicate is true over a range of elements

i.e. over the elements of domain

or

Over the elements in universe of discourse

- There are two types of quantifiers
 1. Universal Quantifier $\forall$: for all
 2. Existential Quantifier $\exists$: there exists



Topic : Universal Quantifier ($\forall$)

$\forall$: for all

$\forall x$: for all 'x' in the domain

or
for all 'x' in the Universe of discourse

Let S : is a sportsman

& domain is the set of all students in our class

$\forall x \{ S(x) \}$: All are sportsman

Truth value of this
predicate formula is the
truth value returned by 'for all' $\forall$

This statement is trying
to state that all
students in our class
are sportsman

it may be true
or
it may be false

- $\forall x \{S(x)\}$ will return false if and only if there is at least one element in the universe of discourse (i.e., in the domain) for which the Predicate $S(x)$ returns false.

Note: If universe of discourse (i.e., domain) happens to be empty, then quantifier " $\forall$ " will always return true, because there is no element for which the underlying Predicate can return false.

→ $\forall x \{S(x)\}$ will return true
if and only if,

- ① domain is Empty
- (or) ② Predicate 'S' returns true for all the elements in the domain

In $\forall x \{S(x)\}$

quantifier

Scope of the quantifier

$\forall x \{P(x) \rightarrow S(x)\}$

Scope of quantifier

H: is human
I: is intelligent

domain: Set of all living things

① $\forall x \{H(x)\} \rightarrow \forall x \{I(x)\}$: if all are human, then all are intelligent
if $\forall x \{H(x)\}$ then $\forall x \{I(x)\}$

② $\forall x \{H(x) \rightarrow I(x)\}$: for all x , if x is human then x is intelligent
|||
All human are intelligent

H.W
Q:- Which of the following Predicate formula is/are valid

$$F_1: [\forall x \{P(x)\} \rightarrow \forall x \{S(x)\}] \rightarrow [\forall x \{P(x) \rightarrow S(x)\}]$$

$$F_2: [\forall x \{P(x) \rightarrow S(x)\}] \rightarrow [\forall x \{P(x)\} \rightarrow \forall x \{S(x)\}]$$

- (a) F_1 only (b) F_2 only
(c) Both F_1 & F_2 (d) Neither F_1 nor F_2

H.W. Consider following two predicate formula

$$F_1: \quad \forall x \{ P(x) \leftrightarrow Q(x) \} \longrightarrow \{ \forall x \{ P(x) \} \leftrightarrow \forall x \{ Q(x) \} \}$$

$$F_2: \quad \{ \forall x \{ P(x) \} \leftrightarrow \forall x \{ Q(x) \} \} \longrightarrow \forall x \{ P(x) \leftrightarrow Q(x) \}$$

Which of the following is true

- (a) Only F_1 is valid
- (b) Only F_2 is valid
- (c) Both F_1 and F_2 are valid
- (d) Neither F_1 nor F_2 is valid



Topic : Existential Quantifier ($\exists$)

$\exists$: There exists
for some / for at least one

$\exists x$: for some 'x' in the domain
(or)
for at least one x in the domain

let P : is a politician
and domain is set of all students in our class.

then $\exists x \{ P(x) \}$: Some (At least one) are politician

Truth value of this
predicate formula is the
Truth value returned by
quantifier " $\exists$ "

It is trying to state that "At least one
student in our class is a politician"

$\exists$ will return true if and only if Predicate under the scope is true for at least one element in the domain

$\therefore$ if domain is empty then

$\exists$ will always return false

$\exists x \{ P(x) \}$

quantifier

scope of quantifier

$\exists x \{ P(x) \rightarrow S(x) \}$

quantifier

scope of quantifier



Topic : Scope of the Quantifier

• The part of the logical expression to which a quantifier can be applied is called the scope of the quantifier.

- A variable whose occurrence is bounded by a quantifier is called a bounded variable. Variables not bounded by any quantifiers are called free variables.

• Generally we represent the scope of the quantifier using brackets or comma otherwise, The Scope of a quantifier is the shortest full sentence/predicate formula which follows it. Everything inside this shortest full sentence is said to be in the scope of the quantifier.

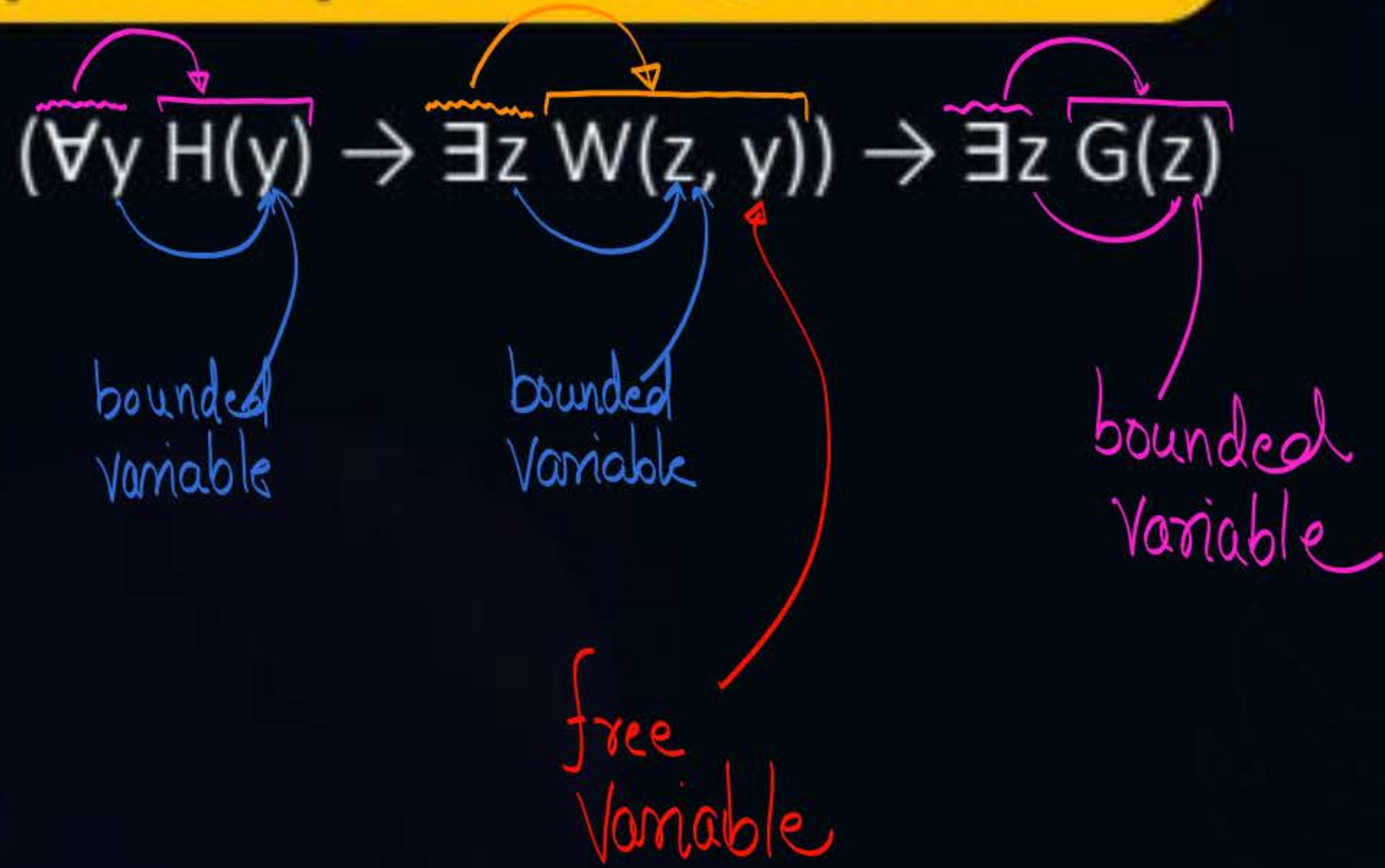


Topic : Scope of the Quantifier

- $(\forall y H(y) \rightarrow \exists z W(z, y)) \rightarrow \exists z G(z)$
- $\exists y(\forall y H(y) \rightarrow \exists z W(z, y)) \rightarrow \exists z G(z)$
- $(\forall y H(y) \rightarrow \exists z G(z)) \rightarrow \exists z W(z, y)$
- $(\forall z H(y) \rightarrow \exists y W(z, y)) \rightarrow \exists z G(z)$
- $(\forall y H(y) \rightarrow \exists z \exists y W(z, y)) \rightarrow \exists z G(z)$
- $(\forall y H(y) \rightarrow \exists z W(z, y)) \rightarrow \exists x G(z)$



Topic : Scope of the Quantifier





Topic : Scope of the Quantifier

$$\exists y (\forall y H(y) \rightarrow \exists z W(z, y)) \rightarrow \exists z G(z)$$

$\exists$ will quantify all free variables 'y' in its scope

bounded by $\forall$

bounded

bounded by $\exists y$

bounded

A single variable in a predicate formula can never be quantified using multiple quantifiers

⇓
If variable is under the scope of multiple quantifiers, then it will be quantified using innermost quantifier



Topic : Scope of the Quantifier

H.W.

$$(\forall y H(y) \rightarrow \exists z G(z)) \rightarrow \exists z W(z, y)$$



H.W.

Topic : Scope of the Quantifier



$$(\forall z H(y) \rightarrow \exists y W(z, y)) \rightarrow \exists z G(z)$$



M.W.

Topic : Scope of the Quantifier



$$(\forall y H(y) \rightarrow \exists z \exists y W(z, y)) \rightarrow \exists z G(z)$$



Topic : Scope of the Quantifier



H.W.

$$(\forall y H(y) \rightarrow \exists z W(z, y)) \rightarrow \exists x G(z)$$



Topic : Note



- Let, $P(x)$: x is true.
 $\therefore \sim P(x)$: x is not true / x is false

	Form	Meaning
1	$\forall x P(x)$	All are true / Not at least one is false
2	$\exists x P(x)$	At least one is True / Not all are false
3	$\sim \forall x P(x)$	Not, all are True / At least one is false
4	$\sim \exists x P(x)$	Not, at least one True / All are false
5	$\forall x \sim P(x)$	All are false / Not at least one is True
6	$\exists x \sim P(x)$	At least one false / Not, all are True
7	$\sim \forall x \sim P(x)$	Not, All are false / At least one is True
8	$\sim \exists x \sim P(x)$	Not, At least one false / All are True



2 mins Summary



✓
Topic

Predicate logic

✓
Topic

Quantifiers

✓
Topic

Scope of quantifiers

THANK - YOU